

ELiRF-UPV at SimpleText 2026: Decoupling Action Prediction and Controlled Generation for Text Simplification

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Proyecto financiado por Secretaría de Estado de Digitalización e Inteligencia Artificial (TSI-100930-2023-9)

Table of Contents

01 **Introduction & Context**
The challenge of simplifying scientific text.

02 **Green AI Philosophy**
Motivation for sustainable architectures.

03 **Two-Stage Pipeline**
Decoupling planning and generation.

04 **Action Prediction**
Sentence-level labeling.

05 **Controlled Generation**
LLM Prompting vs. PEFT Fine-tuning.

06 **Evaluation & Results**
Quantitative and comparative analysis.

07 **Conclusions & Future Work**
Summary of findings and next steps.

ELiRF-UPV — SimpleText 2026

The Challenge: CLEF 2026 SimpleText Track



The Problem

The extensive specialized vocabulary of biomedical literature creates a barrier for non-expert audiences, promoting misinformation.



The Goal

Simplify abstracts and Plain Language Summaries (PLS) from Cochrane systematic reviews.



The Subtasks

Task 1.1

Sentence-level simplification of "shredded" abstracts.

Task 1.2

Document-level simplification to produce cohesive, readable abstracts.

Our Philosophy: Green AI and Sustainability



The Current Paradigm

Recent state-of-the-art approaches heavily rely on prompting massive, private Large Language Models (LLMs) via closed APIs.



Environmental Cost

This "bigger is better" approach incurs high computational, energy, and water costs, while also limiting reproducibility.



Our Proposal

We advocate for using efficient, optimized open models running in local environments to democratize controlled text generation.



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A Decoupled Architecture: Separating Logic from Generation

Two-Module System

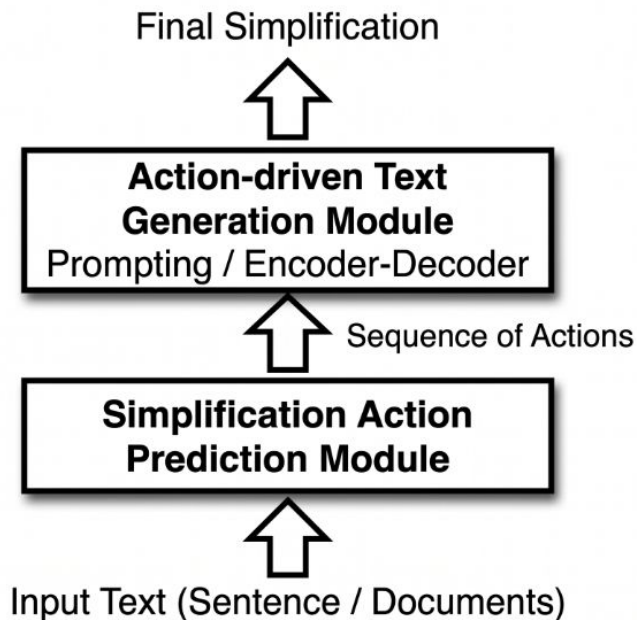
We divide the text simplification task into two distinct stages to maximize control and interpretability.

Stage 1: Action Prediction

An initial module evaluates the source text and predicts specific structural simplification actions.

Stage 2: Generation

A generation module receives these instructions and uses them to explicitly guide the final text production.



SimpleText Simplification Example

Complex Input Text

We identified two non-Cochrane systematic reviews relevant to rehabilitation and the EPOC category of financial arrangements. We excluded one review from the synthesis due to low methodological quality. The other review included 18 primary studies, but only one study reported data relevant to rehabilitation. This study, conducted in a single country (China), compared insured and uninsured children aged birth to 17 years with intellectual disabilities, focusing on the likelihood of receiving at least one rehabilitation service. [...] decision to publish, or preparation of the manuscript. Protocol (2025): DOI 10.23736/S1973-9087.24.08833-6.

Predicted Edit Actions

The sequence of instructional operations applied to restructure the text:

<DELETE> <DELETE> <DELETE>
<REPHRASE> <REPHRASE> <DELETE>
<MERGE_B> + <MERGE_I>
<DELETE> <DELETE> <DELETE>
<DELETE> <COPY> <DELETE>

Simplified Output

We found two systematic reviews (summaries of many studies) about rehabilitation and how it is paid for. We removed one review because it was not done well. The other review looked at 18 studies, but only one study had useful information about rehabilitation. This study took place in China. It compared children from birth to 17 years old with intellectual disabilities. Some children had insurance and some did not. The study looked at whether children received at least one rehabilitation service. [...] We urgently need better studies to help leaders create fair health systems that include rehabilitation.

Stage 1 – Action Prediction (Core Concepts)



Document Context

Built around a shared
ROBERTa-large encoder



Cross-Sentence Attention

Every sentence benefits
dynamically from the context of
the entire surrounding text.

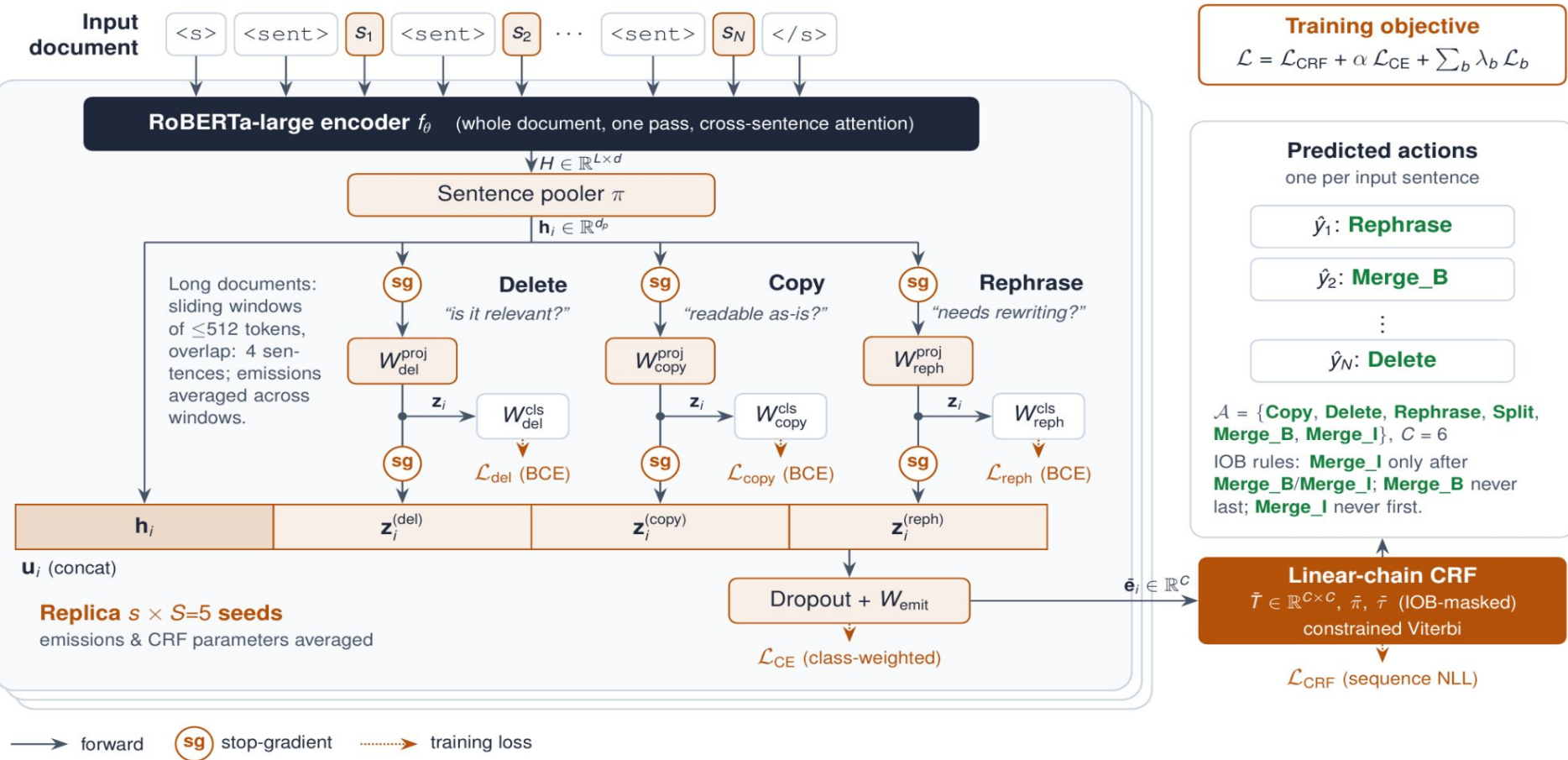


Specialist Branches

Three dedicated binary branches
independently probe orthogonal
semantic axes:

- **Relevance:** Guides the **Delete** action.
- **Readability:** Guides the **Copy** action.
- **Rewriting:** Guides the **Rephrase** action.

Stage 1 - Action Prediction (Architecture)



Stage 2 – Generation Module (Local LLM Prompting Strategy)



Model Selection

We utilize Gemma 4 (31B), a compact open-weight model that enables efficient local inference.



Guidelines

The prompt design is strictly grounded in National Institutes of Health (NIH) guidelines, targeting an 8th-grade US reading level.



Explicit Injection

During inference, each source sentence is sequentially broken down and strictly paired in the prompt with its predicted simplification action.

Stage 2 – Generation Module (Architectural Modification)



Strategy

We introduced 7 special tokens (e.g., <Copy>, <Rephrase>, <Sentence>) to instruct the encoder-decoder models.



The Technical Challenge

Expanding the original vocabulary expands its embedding matrix preventing us from freezing the original pre-trained. Without freezing them, we would be forced to retrain the model entirely.



Proposed Solution

We bypassed this by implementing a **structural separation**. We created an independent, auxiliary embedding matrix dedicated exclusively to the new control tokens, leaving the original matrix safely frozen.

Stage 2 – Generation Module (Dual Conditioning)



Dual Conditioning Strategy

The predicted actions are injected into *both* the encoder and the decoder to maximize control.



Encoder Input

Actions directly precede the complex text (e.g., `<Split> <Sentence> [Complex text...]`) to help the model focus its attention.



Decoder Prefix

These exact same action tags are also passed as an initial prompt to the decoder, strongly reinforcing the expected generation output.

Enforcing Control with DPO



SFT Base Deficit

The model unexpectedly altered text during **<COPY>** actions and failed to completely remove content during **<DELETE>** operations.



Diverse Beam Search

Multiple candidate generations for simplification, selecting the optimal alignment against references via BERT-CRF.



DPO Alignment

The algorithm is applied to penalize bad candidates and reward optimal ones, guaranteeing strict obedience to the tag.

Results & Analysis: The Impact of Action Conditioning

Superior Control:

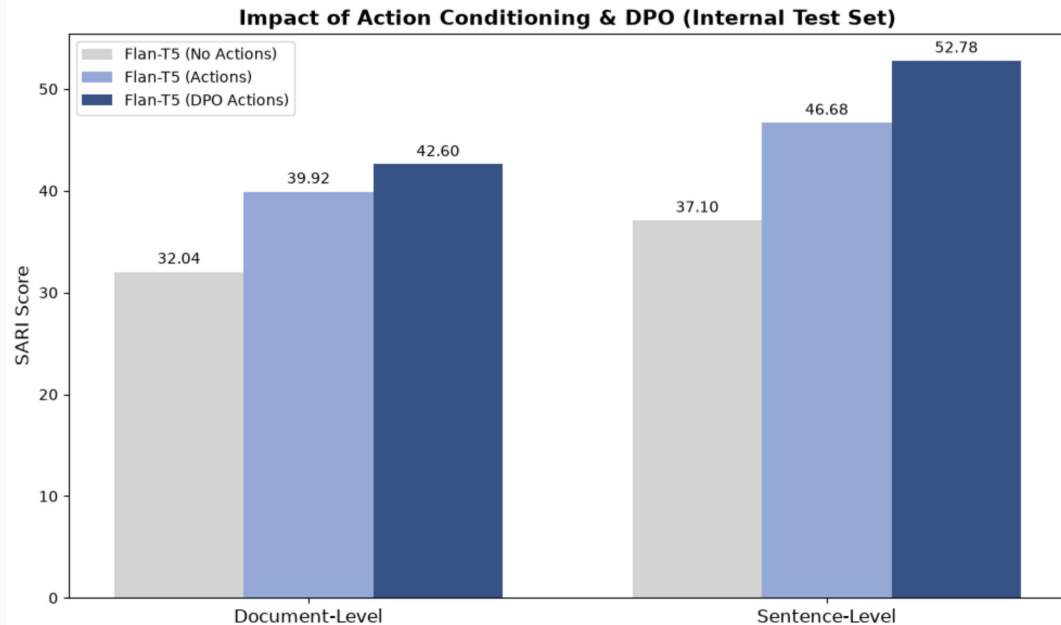
Systems guided by actions (Actions) generalized and performed better than ones without actions (No_Actions).

DPO Success:

DPO training drastically improved the Flan-T5 model.

Subtask Discrepancy:

Sentence-level simplification yielded higher metrics than document-level, driven largely by perfect scores in predicting Copy and Delete actions.



Final 2026 Test Performance

Unexpected Shifts

Performance shifts across all models on the official 2026 test set compared to validation sets.

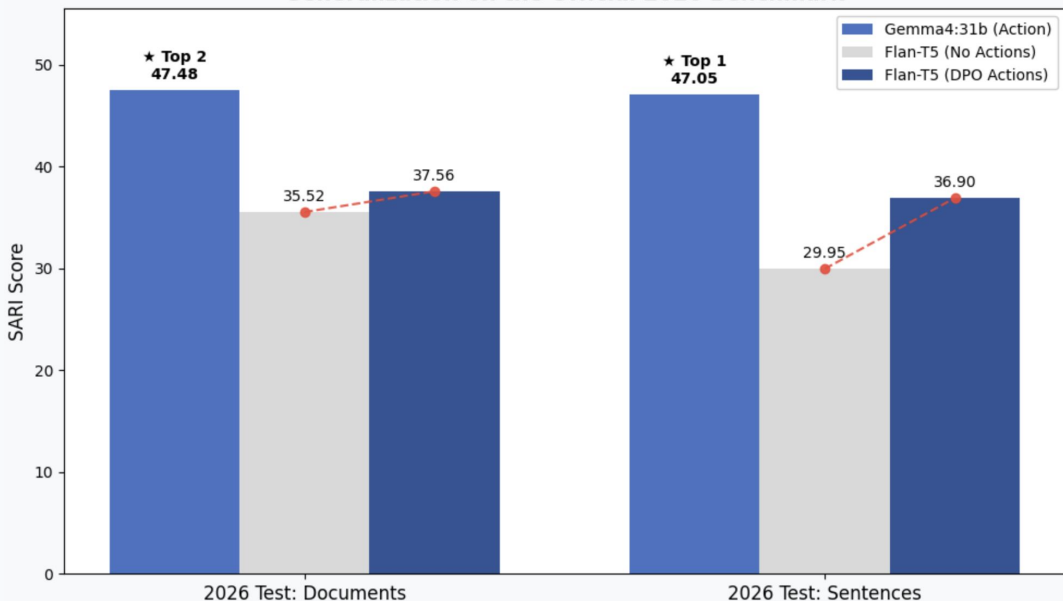
Model Leadership

While Flan-T5 dominated in the internal test, Gemma4 show better results with SARI scores above 47 in both subtasks.

Consistency

Despite the drop, the use of explicit actions and DPO maintained a positive performance gap over baseline models.

Generalization on the Official 2026 Benchmark



Conclusions & Future Work



System Validation

Achieved Top-1 and Top-2 rankings in the aligned Cochrane-auto evaluation.



Sustainable AI

We proved that Green AI guidelines are viable; compact, optimized models can effectively democratize controlled generation.



Future Directions

We plan to evaluate our action injection strategy on newer models and explore a soft integration to help models learn flexibly.

Thanks For Your Attention!

Any Questions?